

Protecting SRv6 Paths

SRv6 steers, IPSec protects

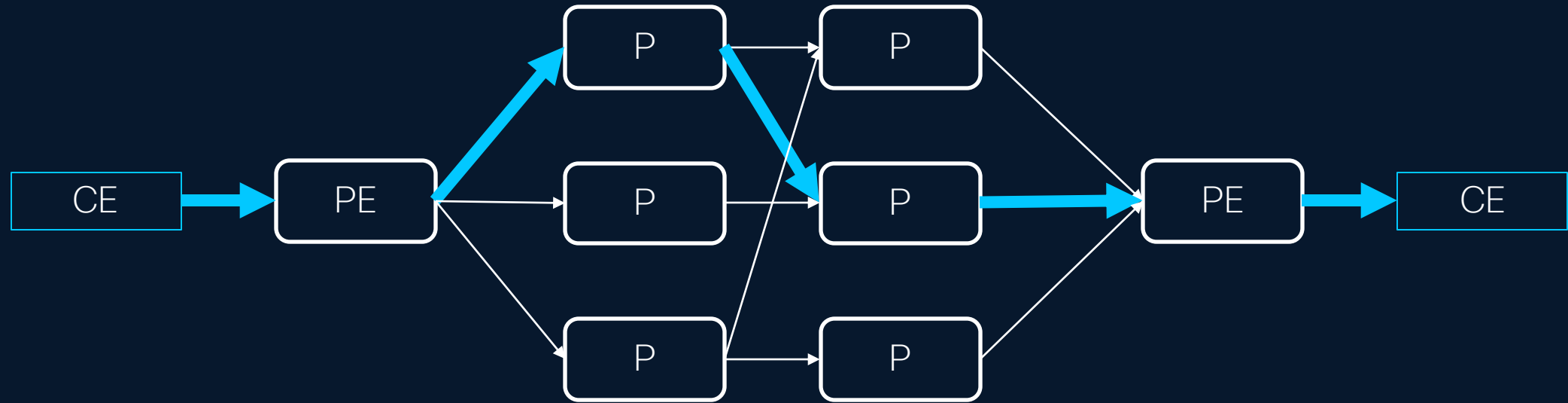
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Agenda

- Problem statement – What we are good at and what is missing
- IPSec overview
- SRv6 + IPSec
- Cisco Silicon One P200 capabilities

SRv6 is good at path control



Path selection



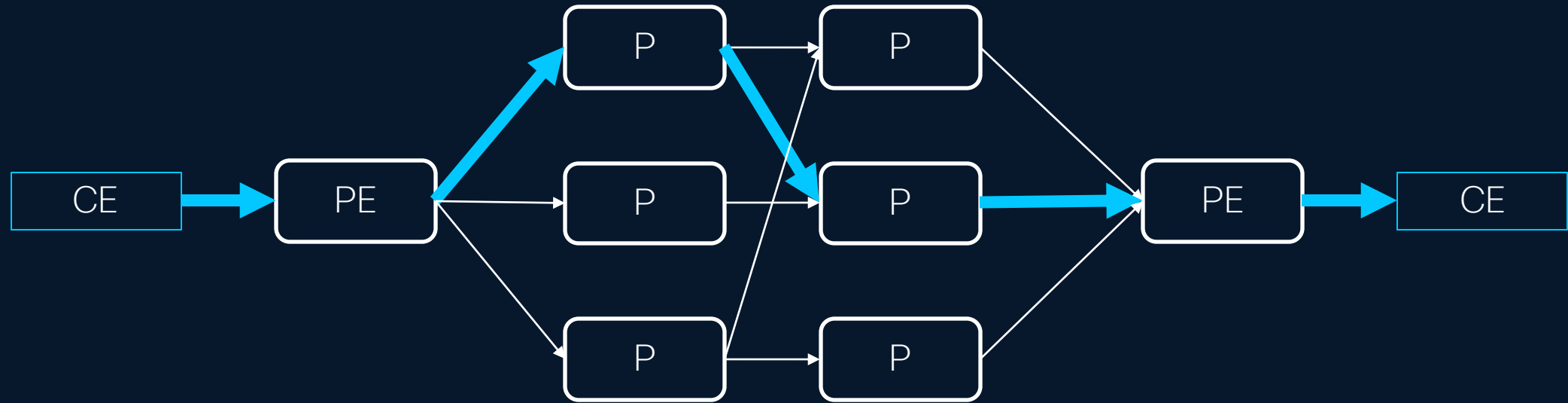
Policy enforcement



Telemetry



SRv6 requires path protection



Confidentiality



Integrity



Authentication

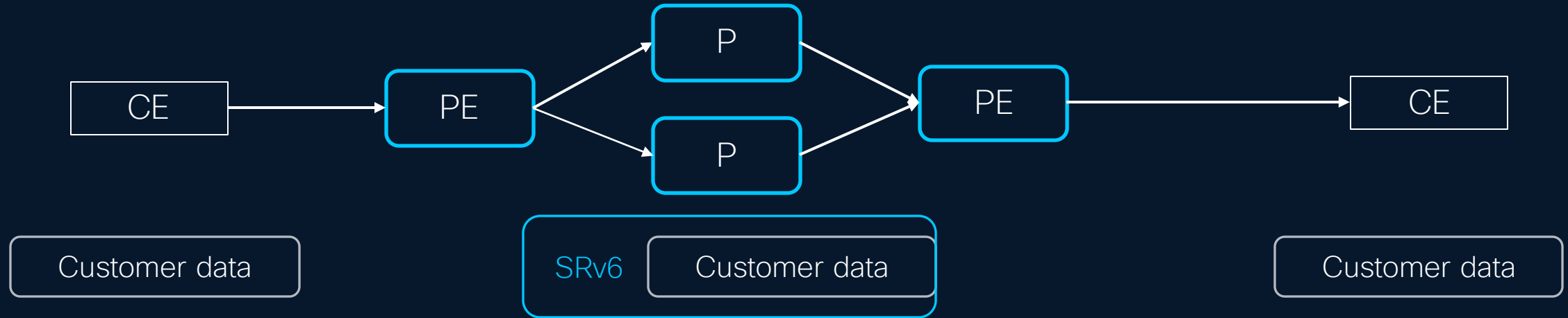


Anti-replay

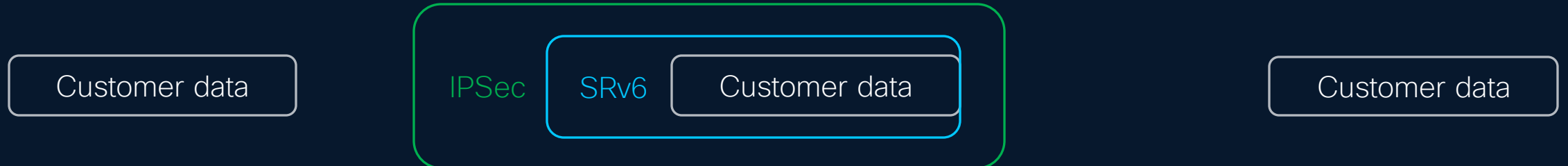


The solution: SRv6 + IPSec

SRv6 is packet encapsulation



IPSec is yet another encapsulation – but with a twist, keeping information in the clear



What is IPSec

A security protocol at the network layer for routed tunnel endpoints

Confidentiality



Transit cannot read it

The inner packet – addresses, payload, inner SIDs – is encrypted. A midpoint with a tap sees outer headers only.

Integrity



Transit cannot change it

A checksum the attacker can recompute is not integrity. IPSec appends an ICV that only the two SA holders can create.

Authentication



It came from the peer PE

A valid ICV means the packet was produced by someone who holds the SA. Forgery of an inner service header fails verification.

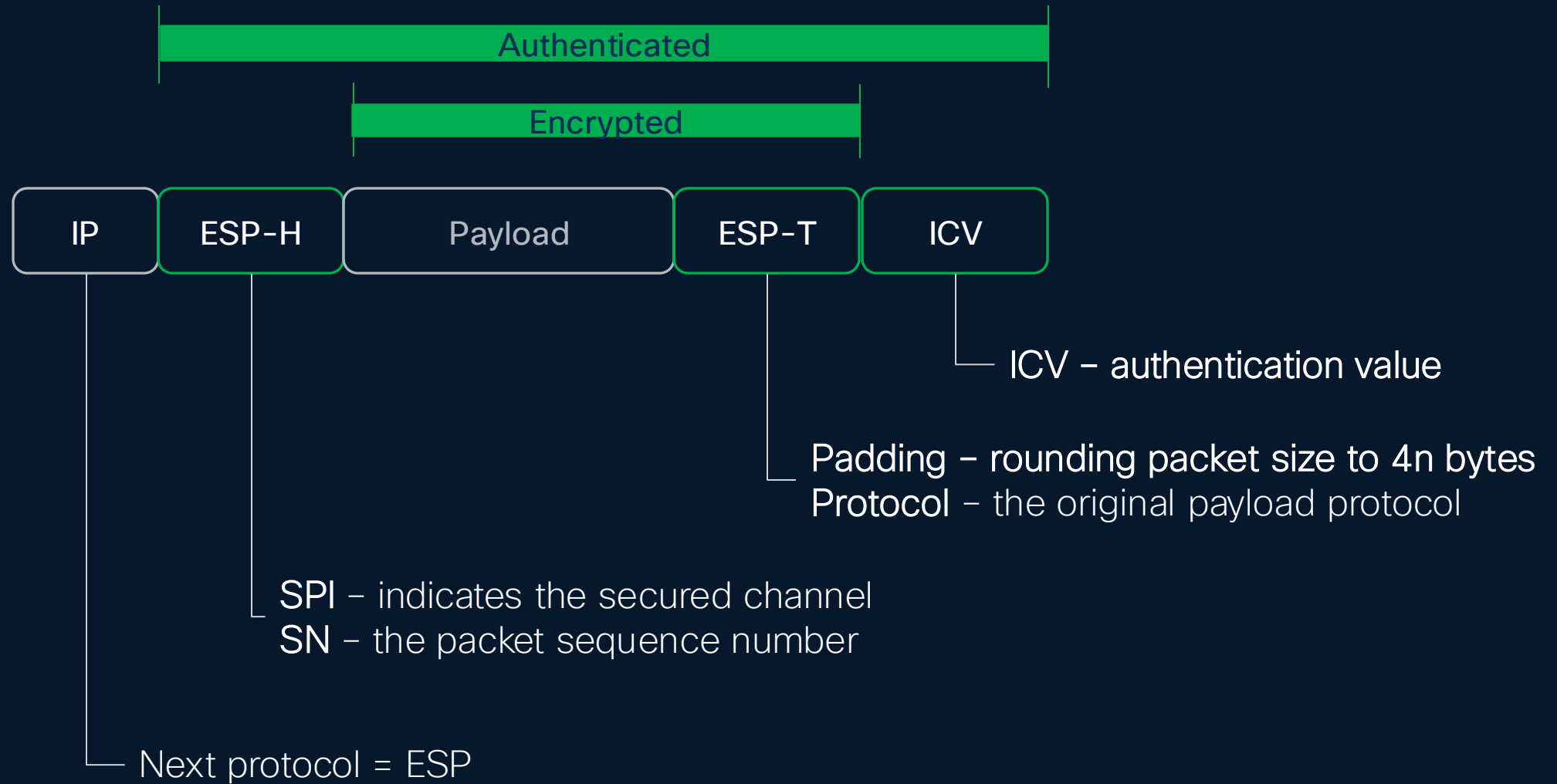
Anti-replay



A capture is not reuseable

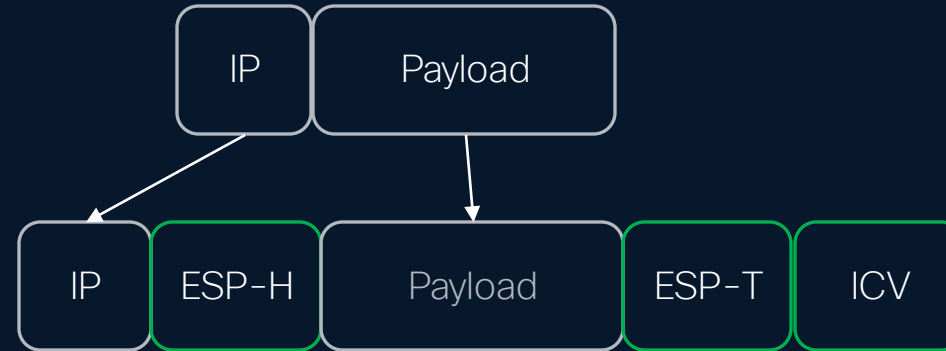
Every packet carries a sequence number. The receiver keeps a window. An old or already received packet is discarded.

IPSec packet format



IPSec modes

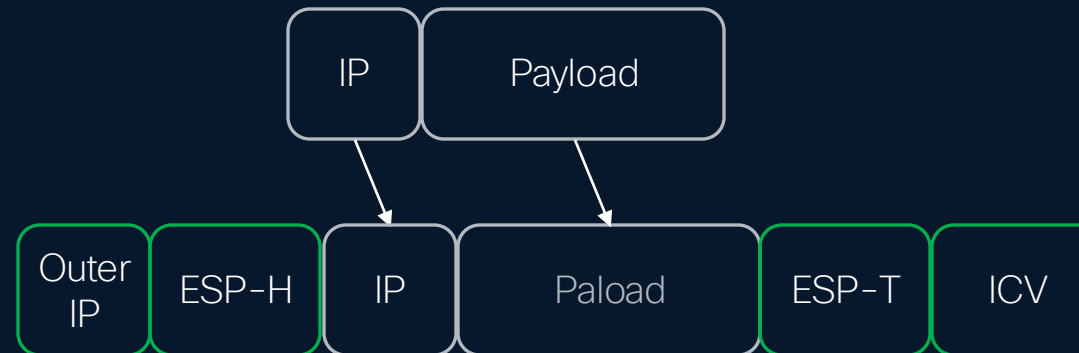
Transport mode



End station to end station protection, VPN

Two secured connections (SAs) between pair of end points

Tunnel mode



Router to router protection

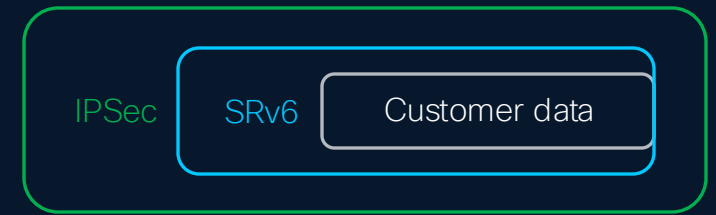
Two secured connections (SAs) between pair of routers

A tunnel aggregates many endpoints

Tunnel mode is what you need for SRv6

SRv6 + IPSec

Option1 – encrypt all



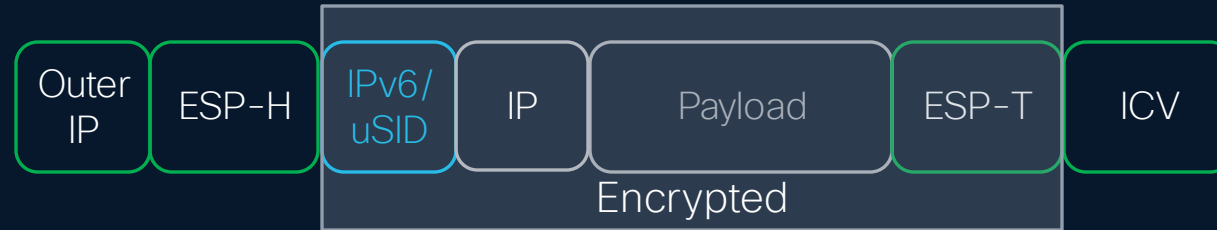
① Customer packet



② Encapsulate customer data with SRv6 packet



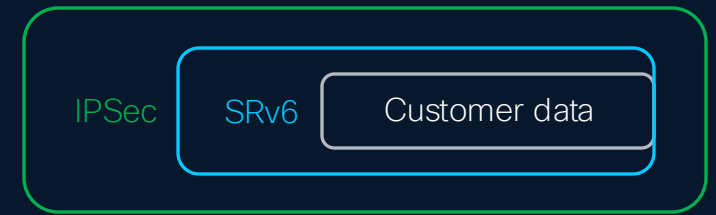
③ Wrap SRv6 with IPSec



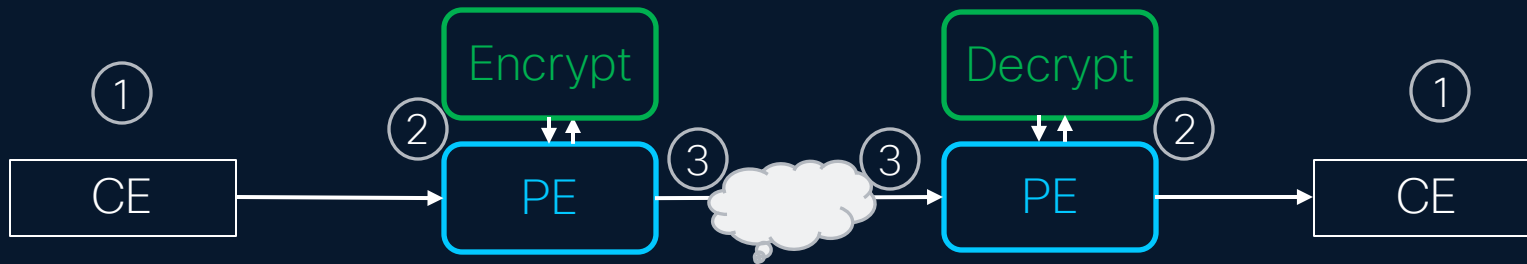
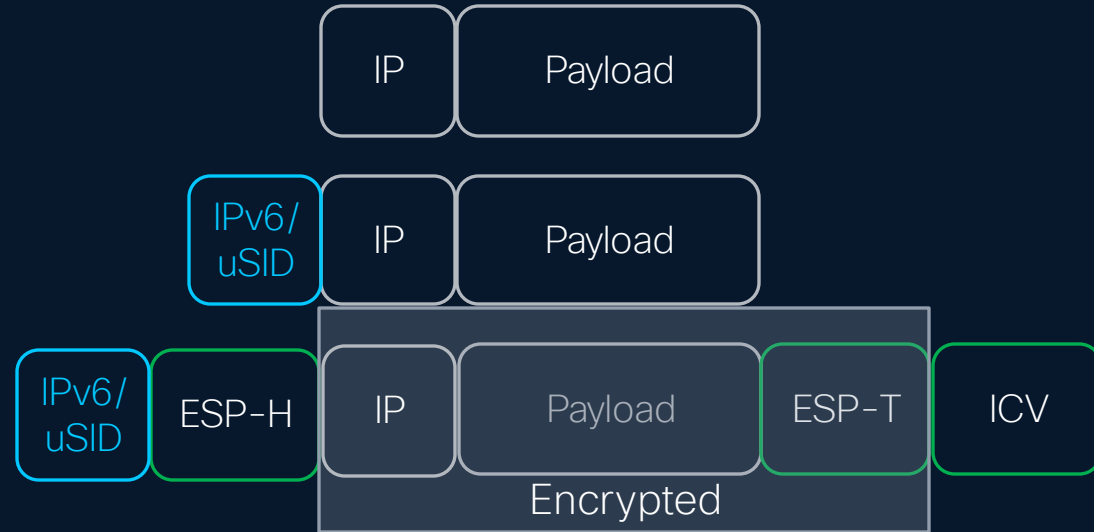
- ⊕ All data is protected and encrypted – nothing is in the clear
- ⊕ Natural and decoupled processing flow: Forwarding ➔ security
- ⊖ Transit nodes cannot inspect the protected uSID list for steering.

SRv6 over IPSec

Option2 – encrypt customer payload



- 1 Customer packet
- 2 Encapsulate customer data with SRv6 packet
- 3 Wrap SRv6 payload with IPSec



- + Customer data is protected and encrypted
- More complex flow – may require appliance (engine-on-a-stick)
- + uSIDs are in the clear and are available for steering at all mid-points and end points

The power of combining SRv6 and IPsec

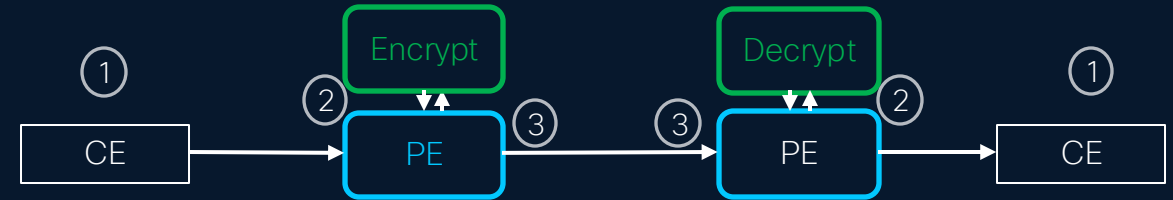
Use case: protect the SRv6 tunnel end-to-end

Security function is not part of the forwarding path and requires

- special hardware engines
- line rate processing of all packet sizes

It may enforce extra appliances and an engine-on-a-stick approach

- Additional interface and extra bandwidth
- May be a bottleneck
- Requires software coordination of SAs
- Different service model
- IEEE 1588 (PTP) 1-step timestamping cannot be supported

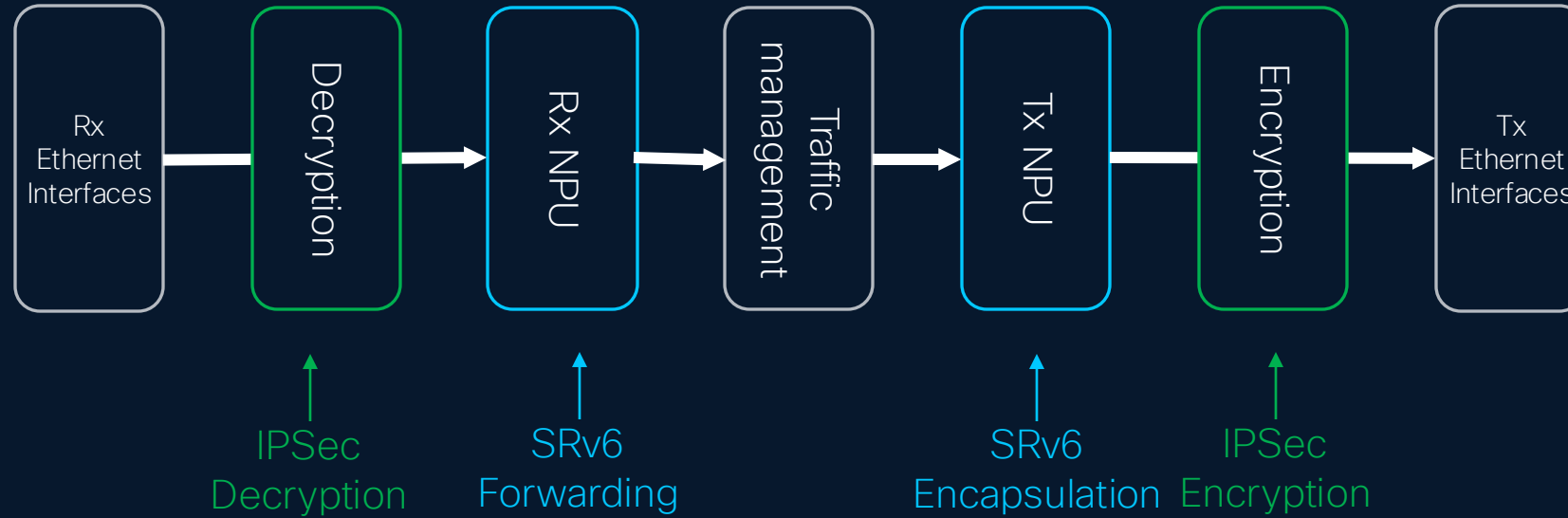


Can the two functions be combined?

Yes - Using Cisco Silicon One P200

Security in P200

Conceptual block diagram



51.2Tbps device

512x100G SerDes

10G,25G,50G,100G,200G,400G, and 800G ports

Standalone and line-card modes

In-line IPsec, MACSec, and ClearTag security engine

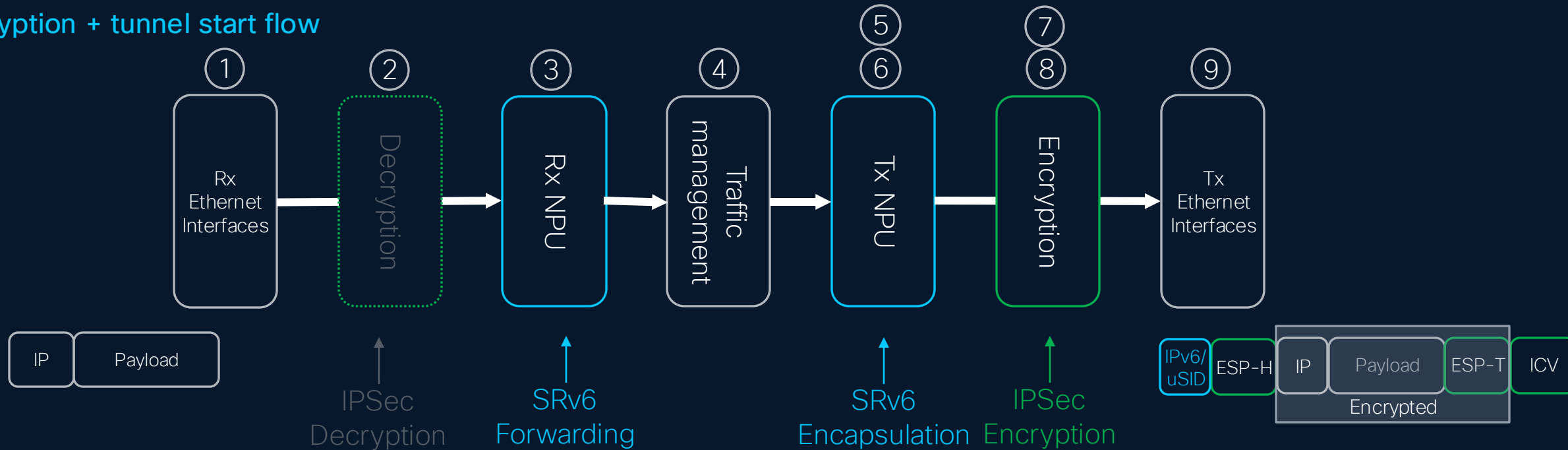
512 Encryption SAs per 800G – total of 32K SAs

576 Decryption SAs per 800G – total of 36K SAs

IEEE 1588 (PTP) timestamping for encrypted packets

SRv6 and IPSec flow in P200

Encryption + tunnel start flow

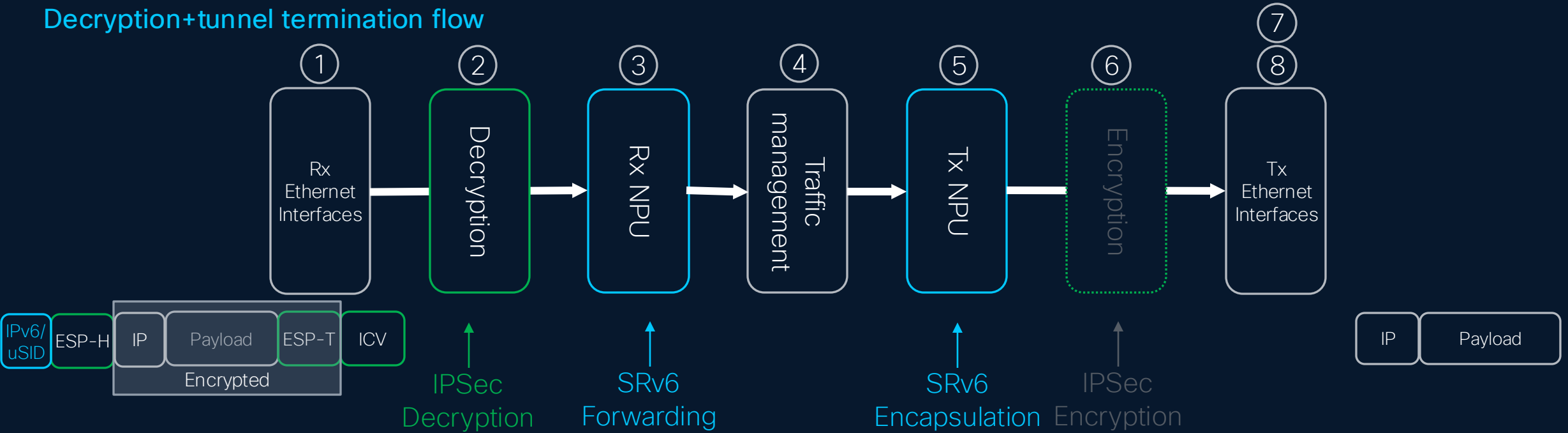


- ① Receive a customer packet
- ② No tunnel termination and no decryption is required
- ③ Forwarding decision to an SRv6 tunnel
- ④ Packet is switched to the target slice
- ⑤ Encapsulate the packet with Srv6 header

- ⑥ If path is protected, resolve SA and send to the encryption engine
- ⑦ Timestmap when applicable
- ⑧ Resolve the encryption key using the SA
Add ESP header,
Encrypt and authenticate the packet payload,
Add ESP trailer and ICV
- ⑨ Append CRC and send the packet out

SRv6 and IPSec flow in P200

Decryption+tunnel termination flow



- 1 Receive a packet
- 2 Classify the packet to a tunnel
Check if the tunnel is set to logical tunnel termination
Resolve the SA
Decrypt the packet (cut-through flow)
Resolve the inner packet type from ESP trailer
Forward the inner packet type to RxNPU
Forward processing-code to RxNPU
- 3 Tunnel termination in RxNPU

- 4 Packet is switched to the target slice
- 5 Decapsulate the SRv6 header
- 6 No SA is forwarded – packet is not encrypted
- 7 Timestmap when applicable
- 8 Append CRC and send the packet out

Key takeaways

SRv6 controls the path; IPSec protects the traffic.

SRv6 provides steering and policy, while IPSec adds confidentiality, integrity, authentication, and anti-replay protection.

Inline crypto keeps security within the normal forwarding path, avoiding the detours, and integration overhead associated with detached crypto engines.

P200 combines SRv6 forwarding and IPSec at high throughput and scale with flexibility to protect only the traffic that requires it.

